

Phase 2 Final Demonstration and Results

****Project:**** Modular Three-Layer RL for Kitchen Manipulation in ROS 2/Gazebo

****Report focus:**** Phase 2 progress after completing the kinematic arm training stack and adding the first Qwen L1-to-RL bridge

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Abstract

Phase 2 advances the project from an operational but incomplete Phase 1 prototype into a working kinematic manipulation research stack with a connected semantic L1 interface. In Phase 1, the custom Gymnasium environment, policy training loop, and deterministic evaluation tools were operational, but the approach-to-docking handoff was unreliable and the VLM component had not started. In Phase 2, the arm-training side was substantially improved: the main controller was simplified to an `Approach -> Finisher` pipeline, and the trained policies now achieve high success across the staged kinematic workspace sweep. In addition, a Qwen-facing MCP bridge was implemented so the L1 model can inspect the scene contract, choose object/source/target slots, resolve a validated `IntentPacket`, and produce an RL-ready high-level skill request.

The project is therefore complete as a kinematic proof-of-concept for the proposed modular architecture. The main remaining work is not conceptual architecture, but migration into a richer Gazebo/ROS visual simulation environment and replacing the current structured scene input with true image-grounded perception.

1. Phase 2 Goal

The original project goal was a modular three-layer robot manipulation system:

```text

L1: semantic task interpretation using VLM/LLM

L2: learned local skill policy

L3: deterministic execution and safety

```

For Phase 2, the practical objective was narrowed to a feasible final demonstration:

- Train the kinematic arm to approach the target region.
- Preserve orientation well enough for insertion / docking behavior.
- Keep the policy interface compatible with later Gazebo and VLM integration.
- Connect the L1 model layer to the trained RL stack through a safe structured interface.

This does not yet claim full physical kitchen manipulation. It demonstrates the core modular

2. System Architecture at Phase 2

The current system has two connected parts.

2.1 Kinematic RL Skill Stack

The manipulation skill stack is implemented in a pure kinematic Gymnasium environment. The action remains a normalized 7-dimensional joint-delta command, and the observation uses the stable dictionary schema developed in Phase 1.

The final practical policy pipeline is:

```
```text
```

```
Approach policy -> Finisher / Dock policy
```

```
```
```

During Phase 1, a separate Dock-Coarse / Bridge path was explored. Phase 2 ablations showed that once the Approach policy learned to produce a sufficiently clean handoff state, Dock-Coarse was no longer useful as a mandatory middle module and could even degrade a good state. The main path was therefore simplified to `Approach -> Finisher`.

2.2 Qwen L1 Bridge

The new L1 bridge exposes a minimal MCP-style tool interface to Qwen:

- `get\_l1\_scene\_context`
- `resolve\_intent\_packet`
- `prepare\_phase1\_skill\_request`

This lets Qwen act as the semantic interpretation layer without violating the L1/L2/L3 boundary. Qwen is allowed to choose task-level fields such as object, source slot, target slot, and constraints. It is not allowed to output raw joint commands, trajectories, torques, or executor-level commands.

The bridge output is a dry-run high-level skill request:

```
```text
```

```
Qwen output -> IntentPacket -> APPROACH -> FINISHER skill request
```

```
```
```

3. Phase 1 Baseline Summary

At the Phase 1 checkpoint, the project was approximately halfway complete:

- The environment, training loop, evaluation harness, and throughput benchmarks were operational.
- Separate approach and dock policies showed measurable progress.
- Switched execution was not reliable.
- The VLM component had not started.

Representative Phase 1 results included:

- Approach policy checkpoints around 71-72% success in the best isolated runs.
- Dock strict success improving from 23% to 44% in representative strict V2 runs.
- Full switched approach+dock success still near 0% in the larger evaluation suite.
- 16-env PPO throughput around 1,762-1,790 aggregate timesteps/s.
- TD3 vectorization scaling from about 272 to 1,658 aggregate timesteps/s from 1 to 8 environments.

These Phase 1 results justified the architecture, but the handoff and VLM pieces were still unresolved.

4. Phase 2 RL Results

The most important Phase 2 change was that the handoff problem was reduced and then simplified. Several intermediate directions were tested: Dock-Coarse, Bridge, readiness classification, acceptance mapping, and finisher adaptation. These were useful diagnostic steps, but the final working simplification was:

```
```text
Approach -> Finisher
```
```

The latest staged workspace sweep shows that the arm can complete the kinematic target task reliably in the trained range:

| Stage | Success Rate | Handoff Position Error | Handoff Orientation Error | Final Position Error |
|-------|--------------|------------------------|---------------------------|----------------------|
| ----: | ----: | ----: | ----: | ----: |
| 0 | 1.00 | 0.50 mm | 0.0073 rad | 1.67 mm 0.0106 rad |
| 1 | 1.00 | 0.62 mm | 0.0099 rad | 1.67 mm 0.0123 rad |
| 2 | 1.00 | 0.85 mm | 0.0119 rad | 1.82 mm 0.0139 rad |
| 3 | 1.00 | 1.20 mm | 0.0138 rad | 2.14 mm 0.0164 rad |
| 4 | 1.00 | 1.71 mm | 0.0150 rad | 2.53 mm 0.0165 rad |

is needed to validate the same behavior in Gazebo or a real execution stack.

5. L1 Qwen-to-RL Demonstration

In Phase 2, the L1 interface was implemented and tested with the local Qwen model setup.

The available local Qwen runtime is:

```
```text
/home/jerry/.openclaw/workspace/qwenvl
/home/jerry/venvs/qwenvl
Qwen/Qwen2.5-VL-7B-Instruct
```
```

For the demo command:

```
```text
Move tray1 from shelf_A1 to shelf_B1 while keeping it level and inserting with a stable
pose.
```
```

Qwen produced a tool-call JSON object:

```
```json
{
 "tool": "resolve_intent_packet",
 "arguments": {
 "object_id": "tray1",
 "source_slot": "shelf_A1",
 "target_slot": "shelf_B1",
 "constraints": {
 "speed_cap": "SLOW"
 }
 }
}
```
```

The MCP bridge resolved this into:

- `IntentPacket` command: `MOVE\_PLATE(shelf\_A1, shelf\_B1)`
- object: `tray1`
- source: `shelf\_A1`
- target: `shelf\_B1`
- high-level skill pipeline: `APPROACH -> FINISHER`

```text

Qwen L1 decision -> IntentPacket -> RL-ready skill request

```

The current bridge is intentionally a dry-run interface. It proves the contract and prevents the LLM from directly controlling the robot.

6. What Was Completed

By the end of Phase 2, the project completed the following:

- A high-throughput kinematic Gymnasium environment for a 7-DoF arm.
- PPO/TD3 training infrastructure and deterministic evaluation.
- A trained kinematic arm skill pipeline capable of approach and stable pose-preserving insertion behavior.
- A simplified `Approach -> Finisher` final policy path after diagnosing and removing unnecessary intermediate modules.
- A Qwen-facing MCP bridge for L1 semantic command interpretation.
- An end-to-end L1 demo artifact showing Qwen producing an RL-ready skill request.
- Regression tests for the new L1 bridge and existing intent-layer validation.

7. Current Limitations

The main limitation is that the final results are still in a pure kinematic environment. This was a deliberate engineering decision because it allowed fast iteration and made the learning problem diagnosable. However, it means the following are not yet solved:

- Full Gazebo physics integration.
- Real camera-based object grounding.
- Contact dynamics, friction, and object interaction.
- ROS controller timing and latency.
- Real robot deployment.

The current Qwen bridge also uses structured scene context. The model can be run as Qwen-VL, but the current final demo does not yet use image-grounded slot/object estimation as the primary source of scene state. The MCP interface is designed so that image-grounded Qwen outputs can be added later without changing the L2/L3 skill contract.

8. Interpretation

The most important Phase 2 result is that the project no longer fails at the same point as Phase 1. In Phase 1, the major unsolved issue was approach-to-dock integration. In Phase 2, the kinematic skill pipeline was trained and simplified enough that the arm can reliably

can now produce a structured command that passes through the L1 validation layer and becomes an RL-ready skill request. This directly supports the original modular research claim: semantic understanding and motor skill learning are separated by a strict interface.

9. Final Conclusion

Phase 2 demonstrates a working modular prototype. The kinematic arm training is effectively complete for the current simplified task: the policy can approach the target, preserve pose, and complete insertion-style finishing behavior with high success in the trained workspace. The L1 Qwen bridge is also operational at the structured-command level, producing validated ``IntentPacket`` objects and dry-run RL skill requests.

The project is not yet a complete Gazebo or real-robot kitchen manipulation system. More time is needed to move from kinematic simulation into a richer visual simulation environment. However, the core modular architecture is now demonstrated: a semantic L1 model can issue a safe structured task command, and the trained RL-based L2/L3 stack can interpret that command as a high-level manipulation skill request.

Evidence Artifacts

- ``artifacts/kinematic_phase1/phase1c/workspace_sweep_workspace_noop_vs_previous_summary_001.json``
- ``artifacts/v5/qwen_l1_demo/l1_to_rl_skill_request_qwen.json``
- ``docs/V5_QWEN_MCP_BRIDGE.md``
- ``docs/PHASE1_APPROACH_DOCK_CLOSEOUT.md``
- ``hrl_ws/src/hrl_trainer/tests/test_v5_qwen_mcp_bridge.py``
- ``hrl_ws/src/hrl_trainer/tests/test_v5_qwen_l1_client.py``